

Local Anesthesia (LA)

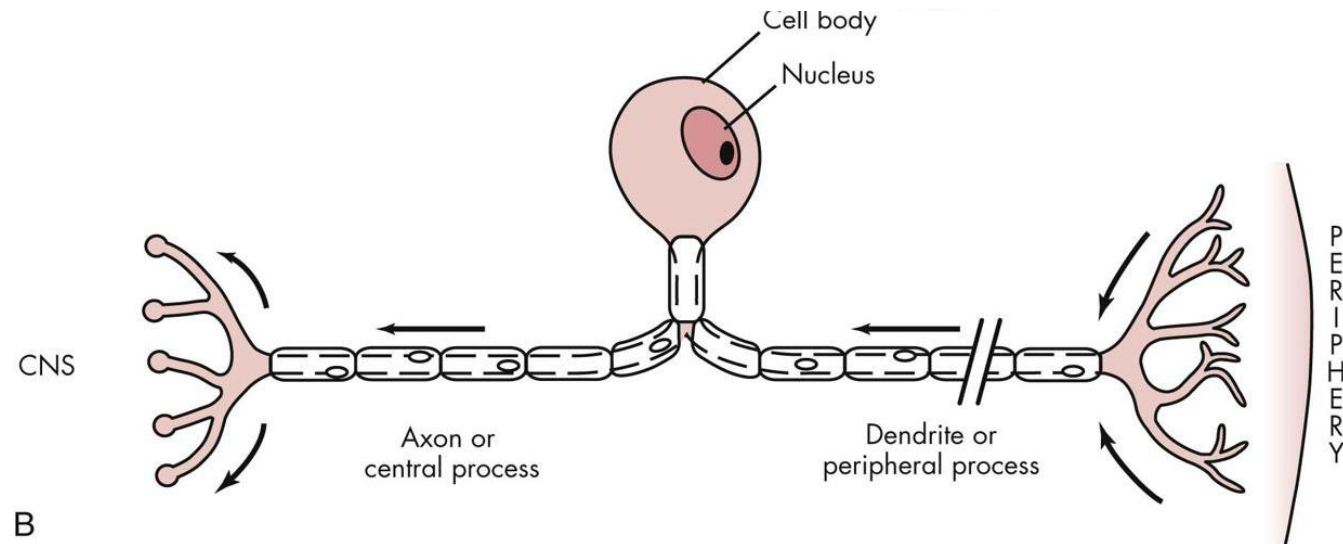
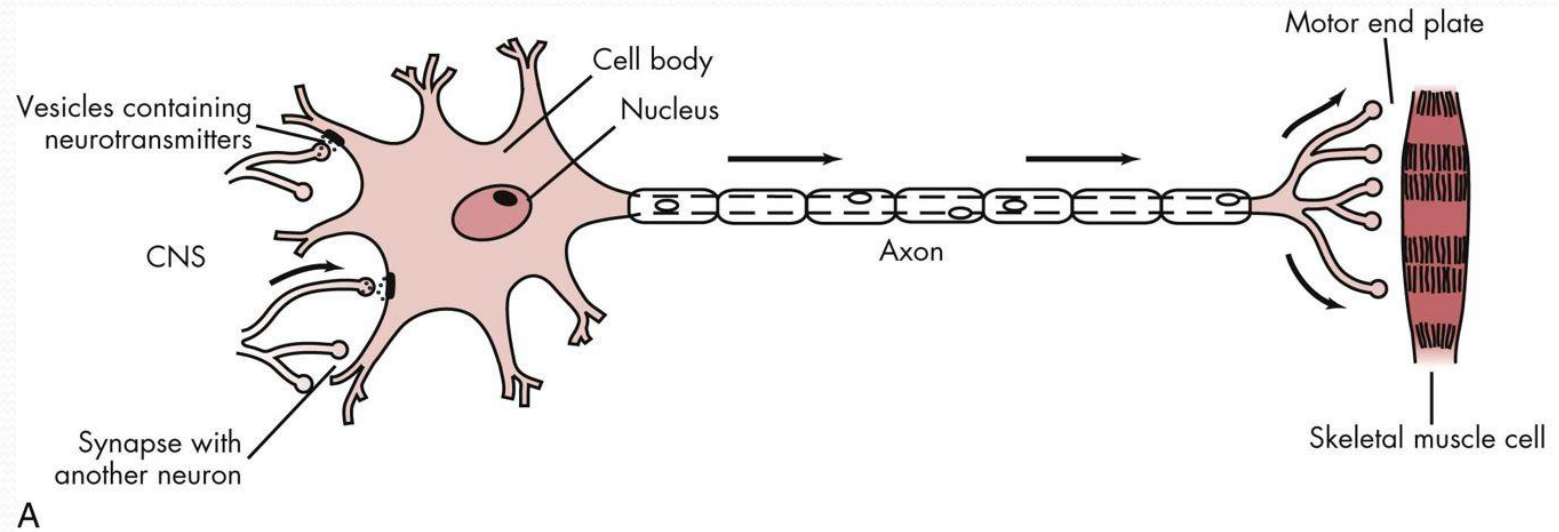
Mode of action

3 BDS Local anesthesia Course

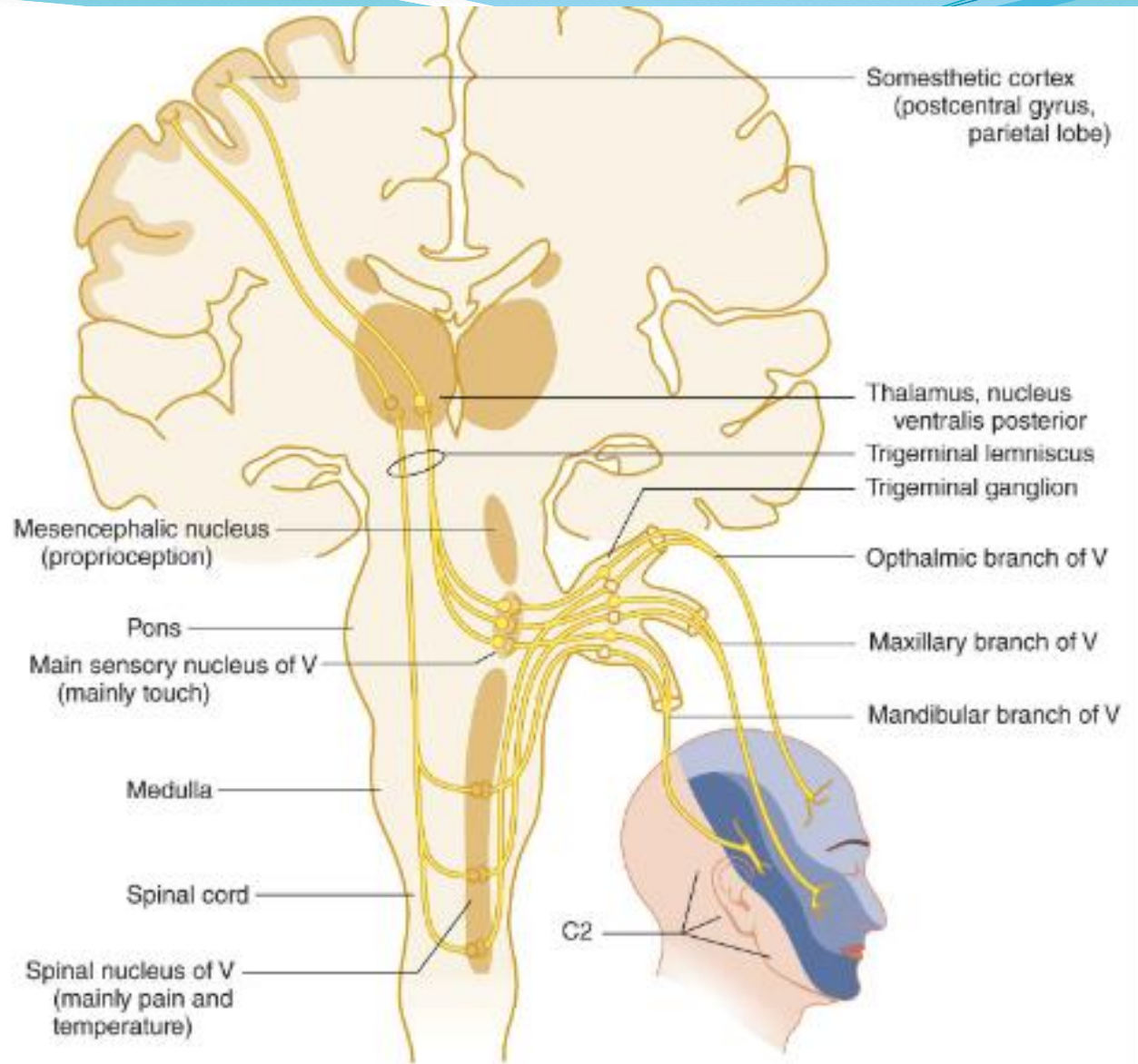
Lecture-3

Dr . Saravanan Ramasamy

Recollection of last lecture



Trigeminal pain pathway



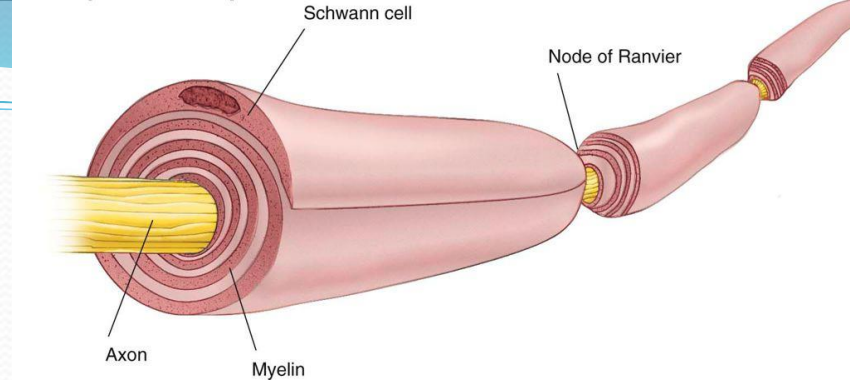
Lecture Learning outcomes

- 1. Describe the nerve anatomy and state its relation to physiology of conduction blockade
- 2. List the molecular components of local anesthetic agents and their role in conduction blockade
- 3. Outline the physiological and pharmacological factors affecting the conduction blockade



Electrophysiology of nerve conduction

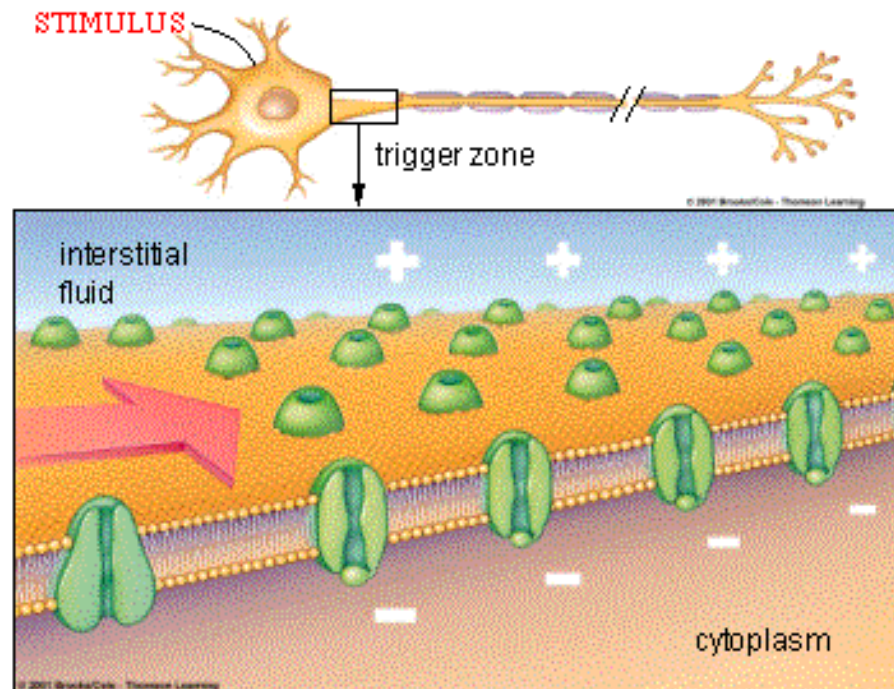
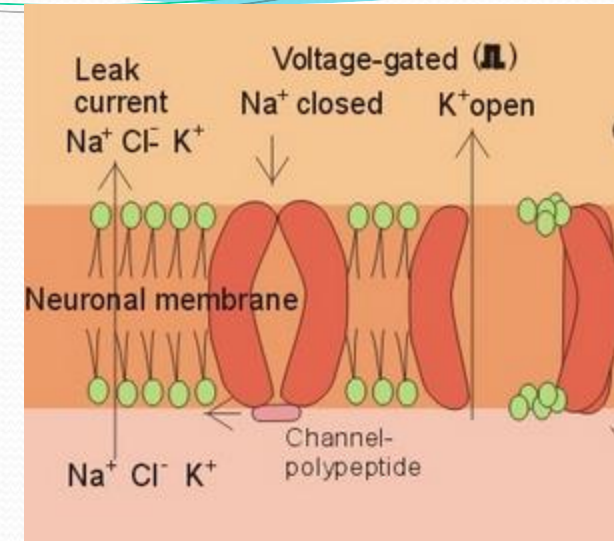
Axon



- The single nerve fiber, the axon, is a long cylinder of neural cytoplasm (axoplasm) encased in a thin sheath, the **nerve membrane, or axolemma**.
- Both sensory nerve excitability and conduction are related to changes developing within the nerve membrane.
- The cell body and the axoplasm are not essential for nerve
- conduction. They are important, for the metabolic support of the membrane

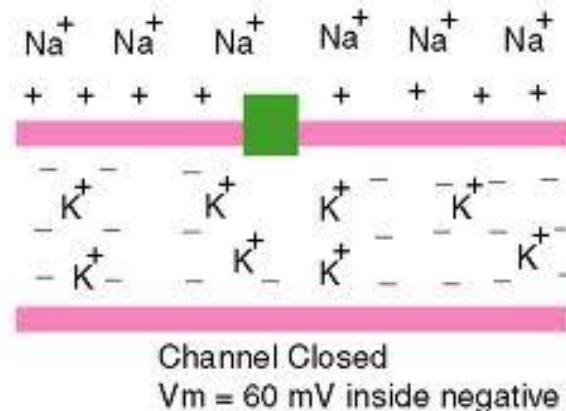
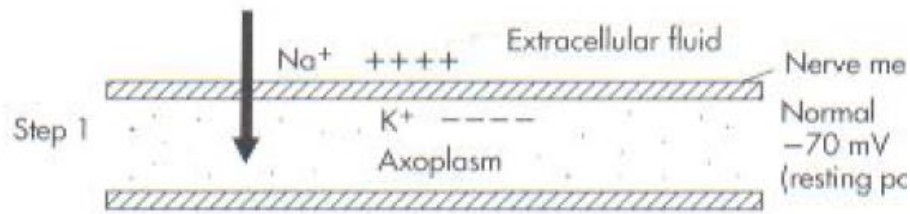
Nerve membrane

- Contains two layers of phospholipids and associated proteins, carbohydrates and lipids
- proteins forms channels that allow selective flow of ions (K^+ , Na^+ , Ca^{++})



Resting stage

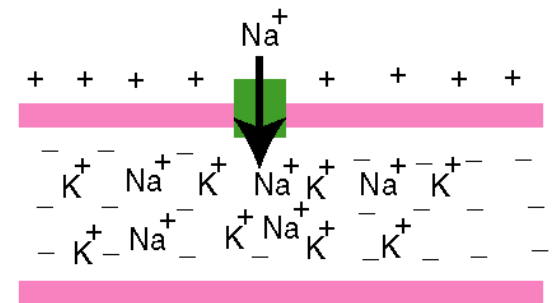
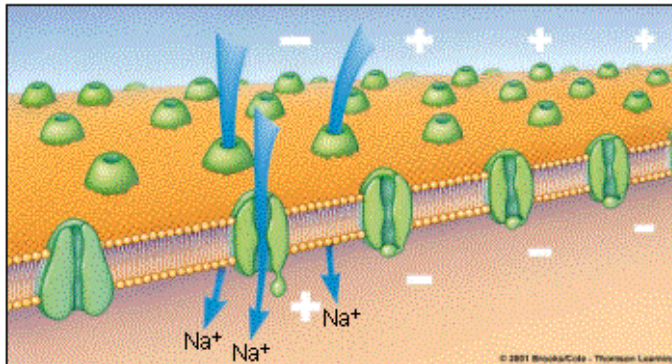
- The interior of nerve membrane is **negative**(-70 mV) in relation to the exterior.
potassium (K^+) remains inside the membrane.
sodium ions (Na^+) remains outside of the nerve membrane because the resting membrane is relatively impermeable to sodium.



Depolarization

- Excitation of a nerve segment leads to **increase in permeability of sodium ions**.
- The rapid influx of **sodium ions to the interior of the nerve** causes a depolarization of the nerve membrane
- At the end of depolarization, the electrical potential of the nerve is reversed (**positive, + 40 mV**)

Action Potential



Channel OPEN
 Na^+ rushes in
 $V_m = 40 \text{ mV}$ inside positive
Close to channel

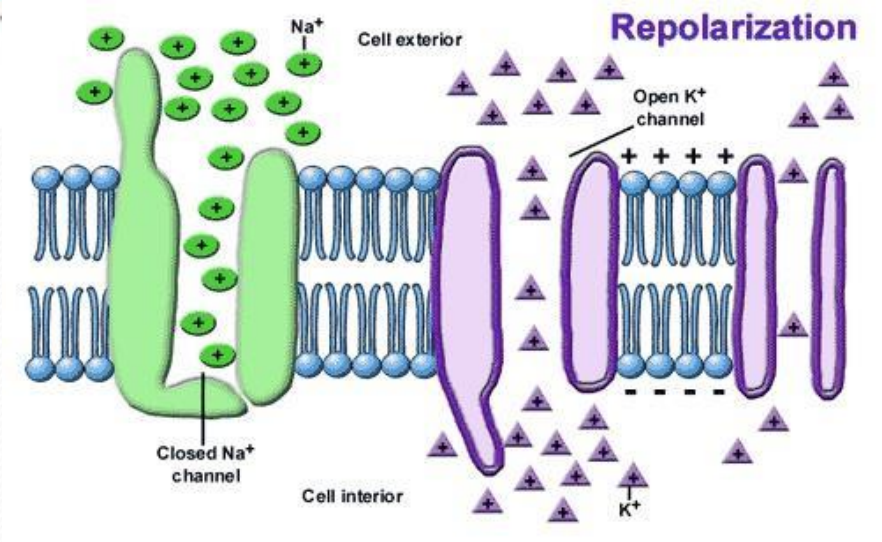
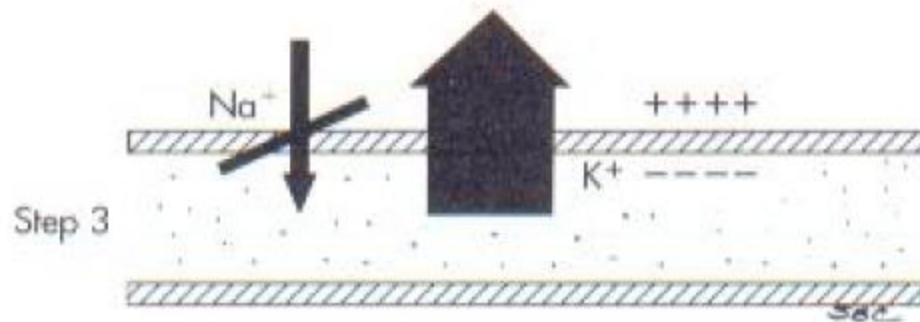
Step 2



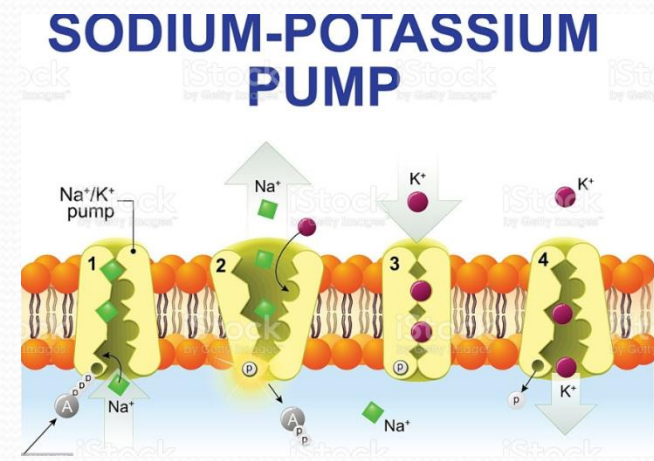
Repolarization

After depolarization, the permeability to potassium increases, resulting an **efflux of potassium** .

This brings the electrical potential back to negative -70mv (repolarization) .

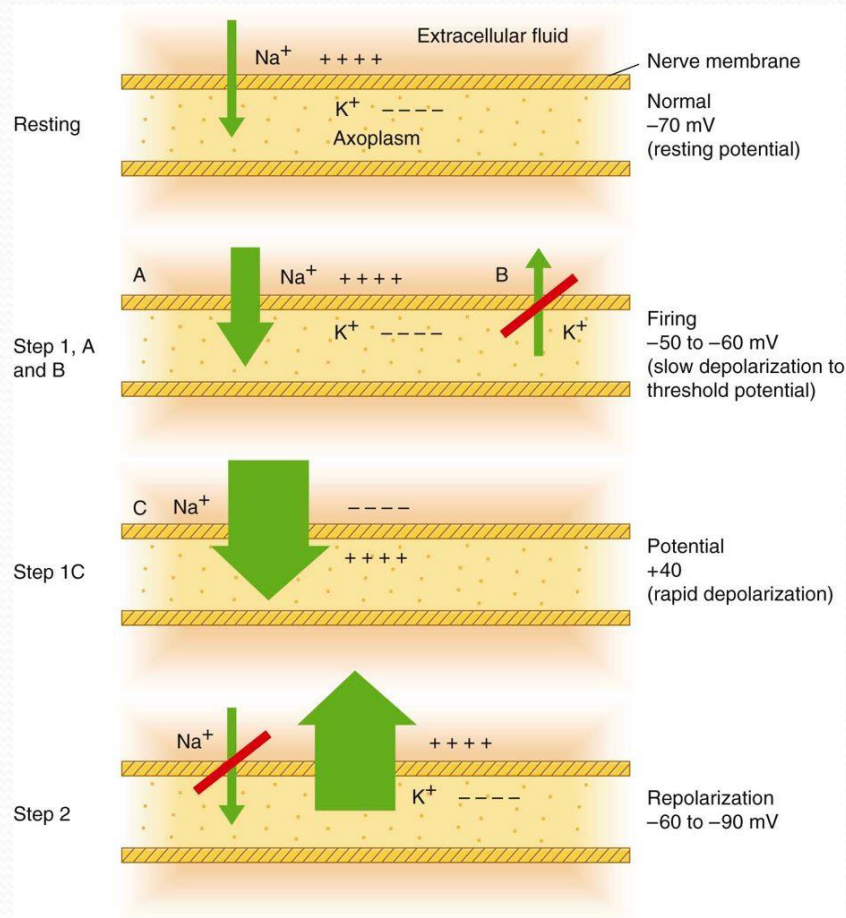


- End of repolarization, Na^+ ions on inside of the membrane, K^+ ions outside the membrane.
- Active transfer of sodium ions outside the membrane occurs through “**sodium-potassium pump**”,
- This energy coming from oxidative metabolism
- of adenosine triphosphate (ATP).
- Sodium ions are moved out of the nerve cell, potassium ions are transported into the cell. The resting potential is re-established.



Summarizing Action potential generation

Resting stage

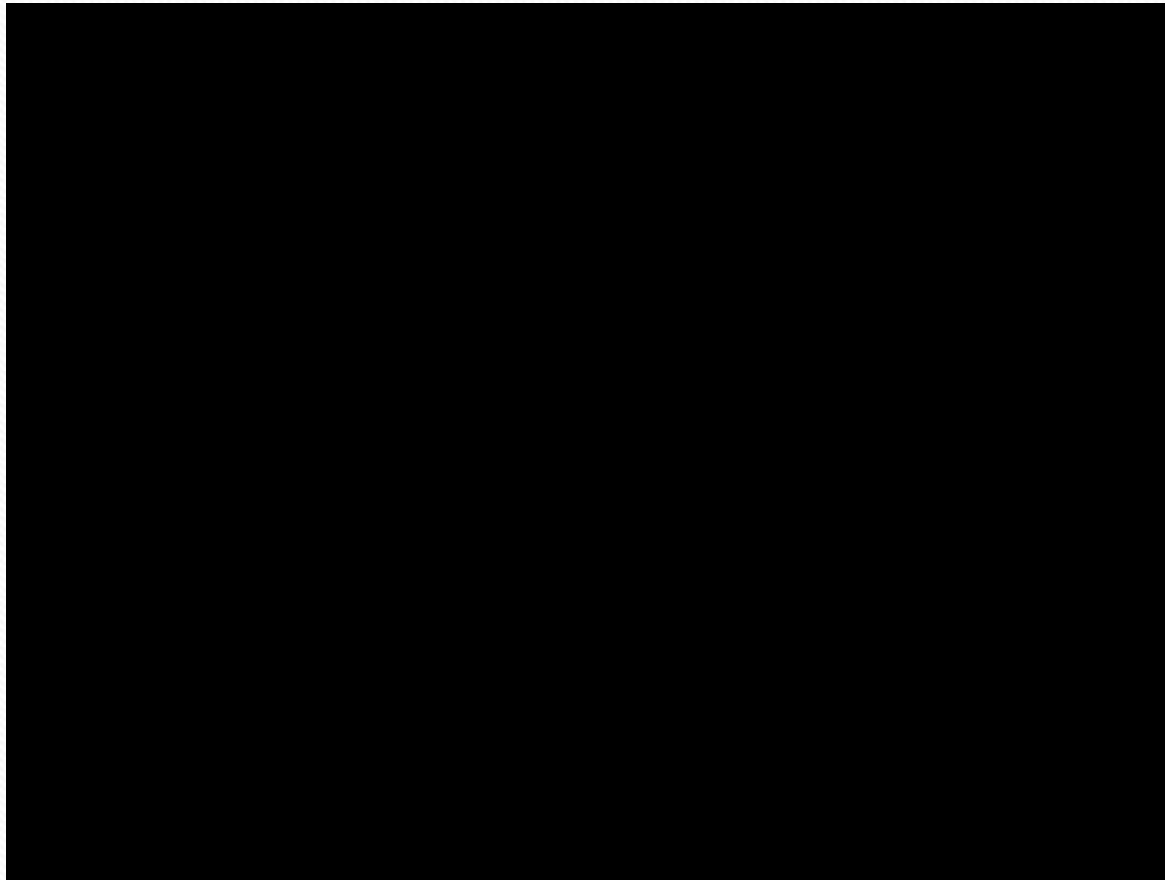


Depolarization

Repolarization

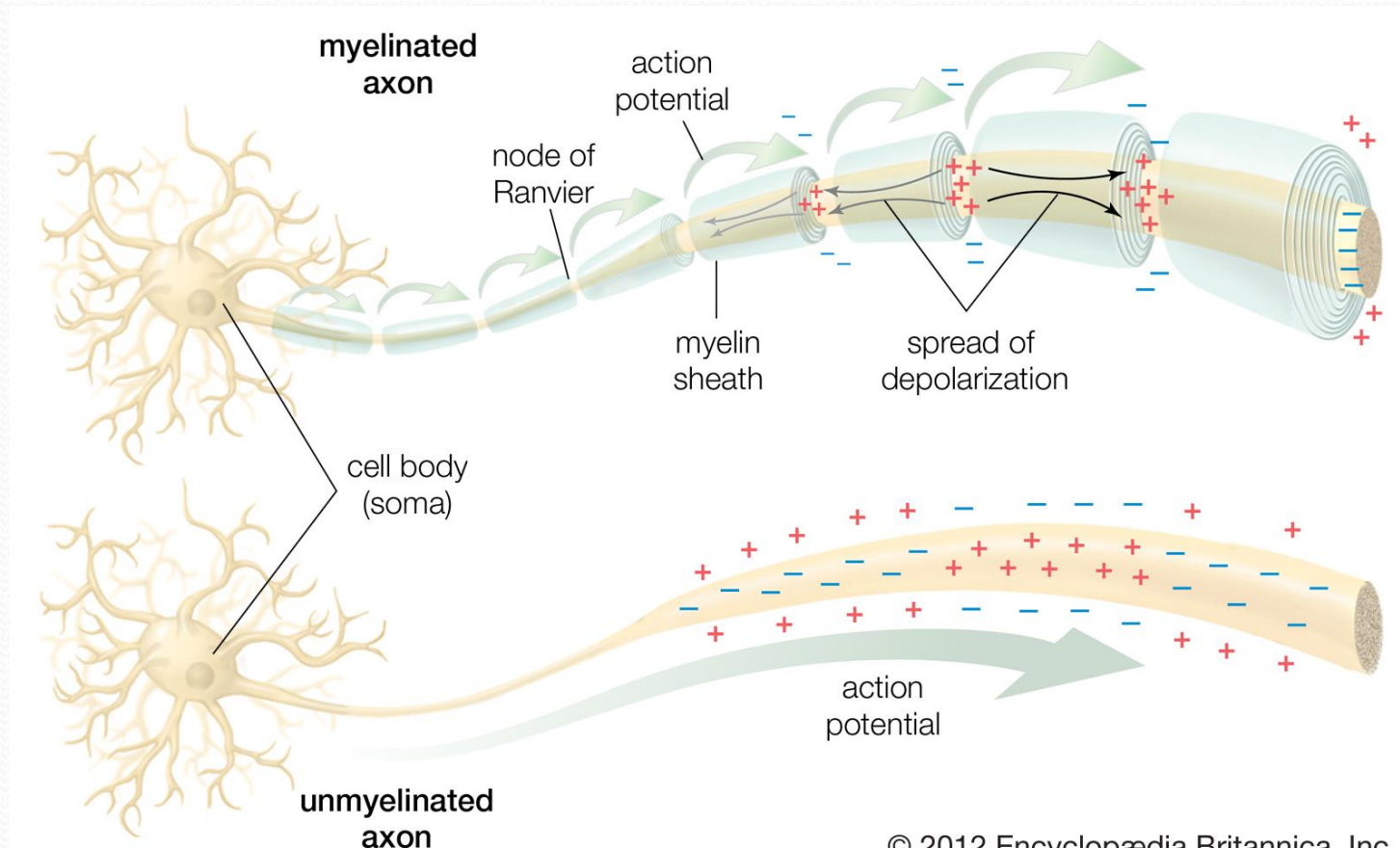
"Nerve Impulse Molecular Mechanism [3D Animation]" on
YouTube

<https://youtu.be/fHRC8SlLcH0?si=cYlQyFg6MphXGqwO>



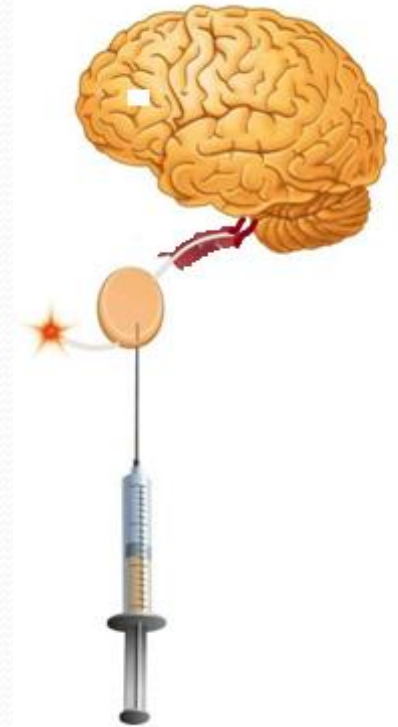
So how fast this process?

- The entire process requires **1 millisecond**
- depolarization takes **0.3 msec** & repolarization takes **0.7 msec**



So how does the LA work

- The concept behind the actions of local anesthetics is simple: They prevent both the generation and the conduction of a nerve impulse.
- In effect, local anesthetics set up a **chemical roadblock** between the source of the impulse (e.g., the scalpel incision in soft tissues) and the brain.
- Therefore the impulse is aborted, prevented from reaching the brain, cannot be interpreted by the patient as pain.



So how does the LA work

- The primary action of local anesthetics is to **decrease the permeability of sodium ions(Na^+)** into the nerve membrane.

Currently two theories of LA are mostly accepted

- **Membrane expansion theory**
- **Specific receptor theory**

- **Old theories which are outdated**

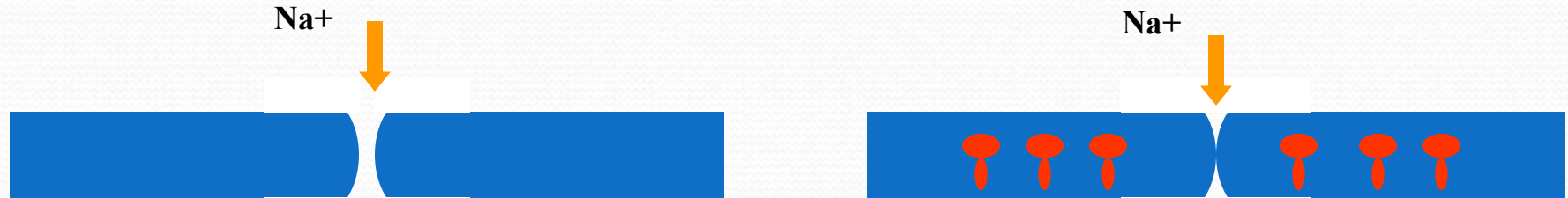
- - Acetylcholine theory
- Calcium displacement theory
- Surface charge theory

Membrane expansion theory

- States that Local anesthetic molecules diffuse to the hydrophobic regions of nerve membrane
- expanding the membrane
- resulting in decreases diameter of sodium channels
- leads to inhibition of sodium influx and neural excitation

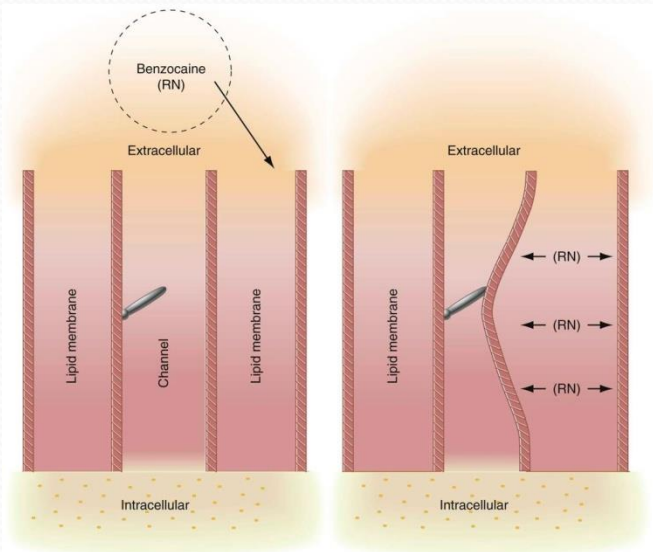
example; benzocaine is a highly lipid soluble LA, used as a topical anesthetic.

Membrane Expansion Theory



**LA Agents
(Benzocaine)**

**Increases lateral pressure
within nerve membrane**



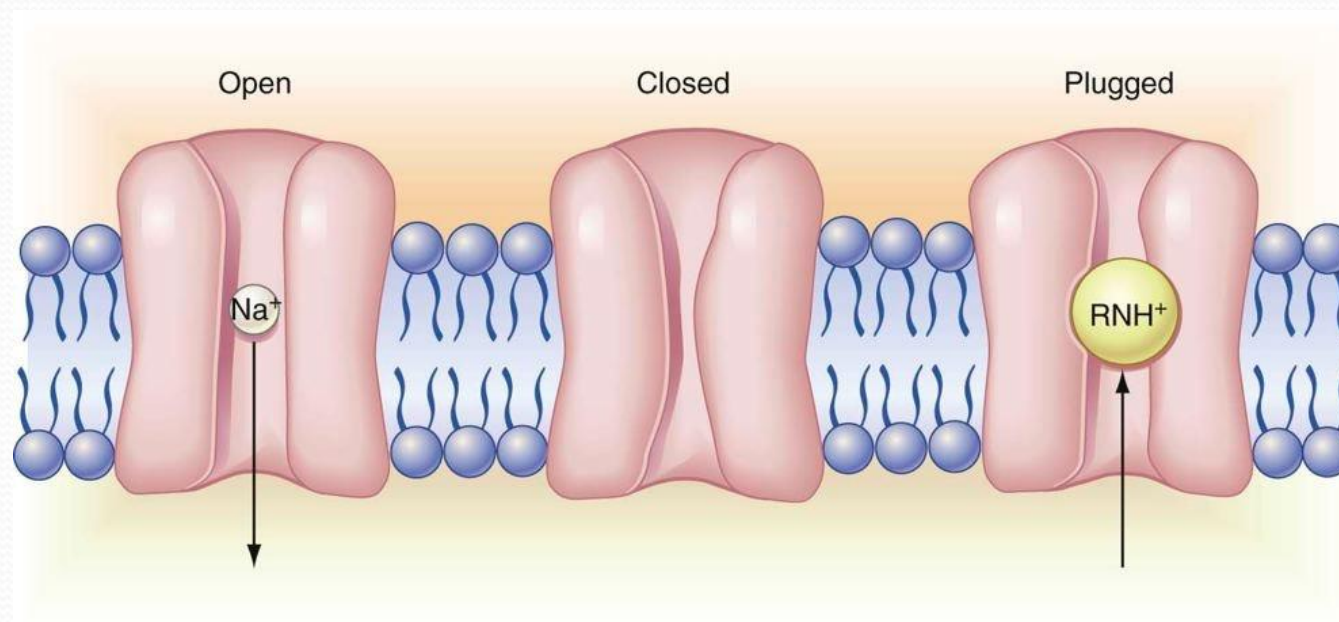
**Constriction of sodium
channels through which Na
passes during
depolarization**

**Interrupt neural
excitation**

Specific Receptor Theory

Local anesthetics act by binding to specific receptors on the sodium Channels

So the permeability of sodium ions is decreased and nerve conduction is Interrupted.

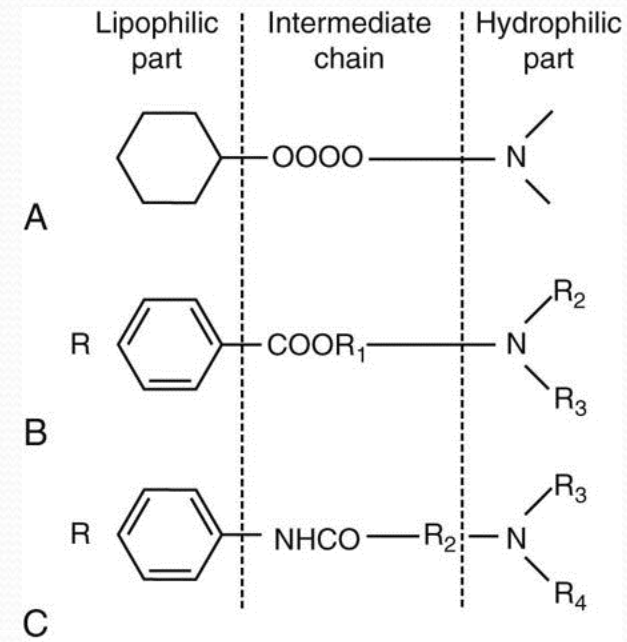


Mechanism of action of LAs:

1. Binding of the LA molecule to Na⁺ channel receptor site , which produces
2. **Blockade of the sodium channel**, and
3. Decrease in Na⁺ conductance, which leads to..
3. Depression of the rate of electrical depolarization and
4. Failure to achieve the threshold potential level,
- 5.No depolarization
6. **Conduction blockade.**

Local Anaesthetic molecules

- Most of Local Anesthetic Agents are chemically made of **Tertiary Amines**
- Local Anesthetic agents are **Ampipathic in nature**
Ampipathic = Lipophilic and Hydrophilic parts located at opposite end.
- Lipophilic and Hydrophilic part are linked by a intermediate hydrocarbon chain.
- This intermediate hydrocarbon chain can either be **Ester (Fig B) or Amide (Fig C)**



Hydrophilic Part

Chemical configuration of local anesthetics.

Aromatic residue	Intermediate chain	Amino terminus	Aromatic residue	Intermediate chain	Amino terminus
ESTERS			AMIDES		
Procaine			Lidocaine		
Propoxycaine			Etidocaine		
Tetracaine			Mepivacaine		
Cocaine			Bupivacaine		
Benzocaine			Prilocaine		
Dyclonine*			Articaine		

*Erythronine is a ketone.

• Esters

• Amides

The main difference between amides and esters

- The nature of the linkage is important in defining several properties of the local anesthetic, including the **basic mode of biotransformation**.
- Ester-linked local anesthetics (e.g., procaine) are readily hydrolyzed in plasma.
- Amide-linked local anesthetics (e.g., lidocaine) are relatively resistant to hydrolysis. Amides are metabolized in liver.
- So Amides are stronger anesthetics, Have longer duration of anesthesia, Currently All clinically useful local anesthetics are Amides



All Local Anesthetic Agents when prepared are .

- Weak basic compounds
- Poorly soluble in water
- Unstable on exposure to air

To alter the above properties, all local anesthetic agents are treated with acids to form salt.

Thus local anesthetics used for injection are dispensed as acid salts, most commonly hydrochloride salt.

(e.g, Lidocaine HCL, Articaine HCL) dissolved in sterile water.

END OF LECTURE 1

References :

Page No 11-23, Chapter-1 Neurophysiology

Hand Book of Local Anesthesia by
Stanley .F. Malamed
7th Edition

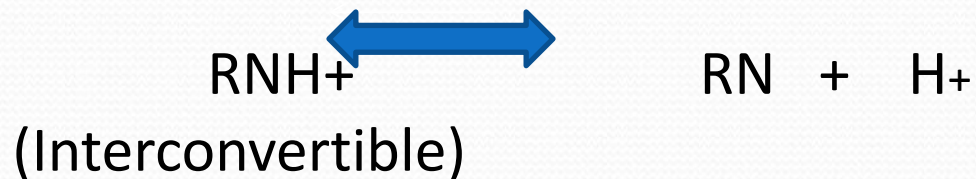
- "Nerve Impulse Molecular Mechanism [3D Animation]" on YouTube
<https://youtu.be/fHRC8SILcHo?si=cYlQyFg6MphXGqwO>

Dissociation of Local Anesthetic Agents

Local Anesthetic Agents when dissolved in water represent two forms

- Uncharged **Base form**- RN (**Lipophilic**)

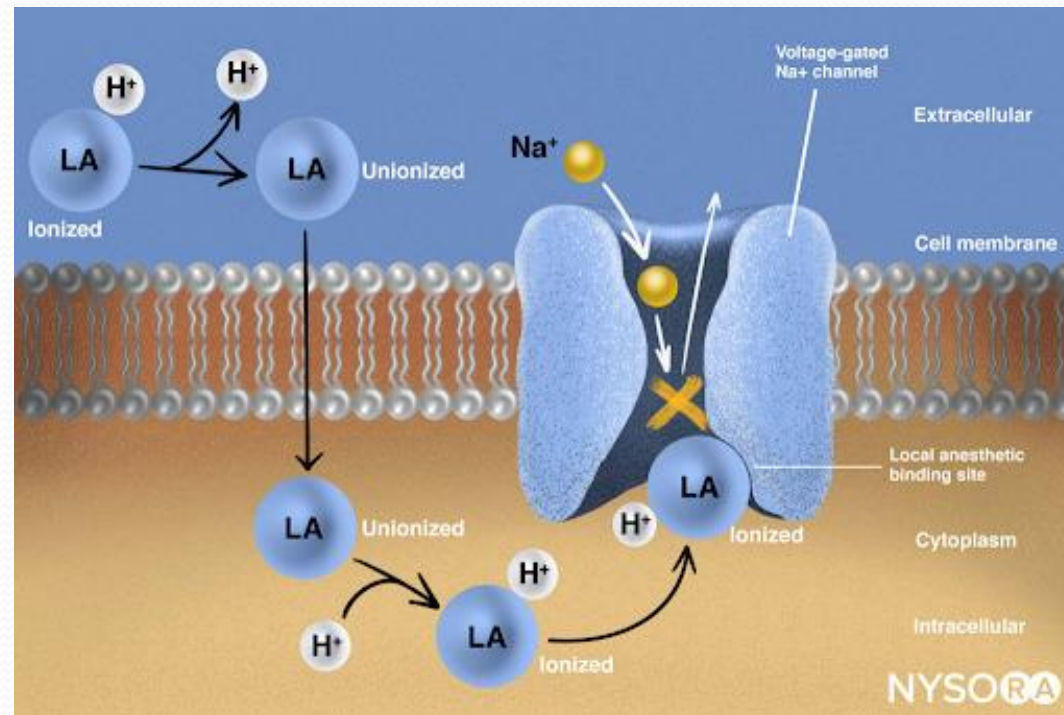
-Positively charged **Cation form**- RNH⁺ (**Hydrophilic**)



RN is uncharged ,**Lipophilic** and can only Cross Nerve membrane (Bilipid Layer)

RNH⁺ Cation form is **Hydrophilic**- Only can attach to the receptors in the sodium channel and cause conduction block.

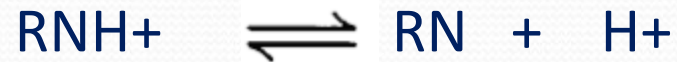
Site of receptors for RNH⁺ Cations



Dissociation Constant (pKa)

Is a measure of affinity of a molecule for Hydrogen ions(H⁺)

Dissociation Constant (**pKa**) of a local Anesthetic Agent is that pH in which **Base form** and **Cation form** are equal in number.



Base form and **Cation form** are interconvertible.

Example- Drug with pKa 7.7

This means at pH 7.7 , there are equal number of RNH⁺ & RN form molecules of that particular drug.

Dissociation Constant (pKa) of Various local anesthetic agents range between 7.5 -10

The relative proportion of the each form is dependent on Ph of solution

In Low pH (Acidic pH) there are high concentration of hydrogen H^+ ions .That means the above formula shifts towards left , i.e **more cation form** is present than Base form in the anesthetic



In Higher pH (Basic pH) there are less H^+ ions .That means the above formula shifts towards right, i.e more Base form is present than Cation form.



Henderson – Hesselbalch Equation

This equation calculates the number of Cation form and Base form molecules of a Local Anesthetic Agents

Requirements for this equation are

- Dissociation constant (pKa) of the Local Anesthetic agent
- Ph of the surrounding tissue in which solution is deposited then.

Distribution of the drug in Cation ion form and Base form at **Body tissue ph 7.4** (calculated by Henderson Hasselbalch Equation)

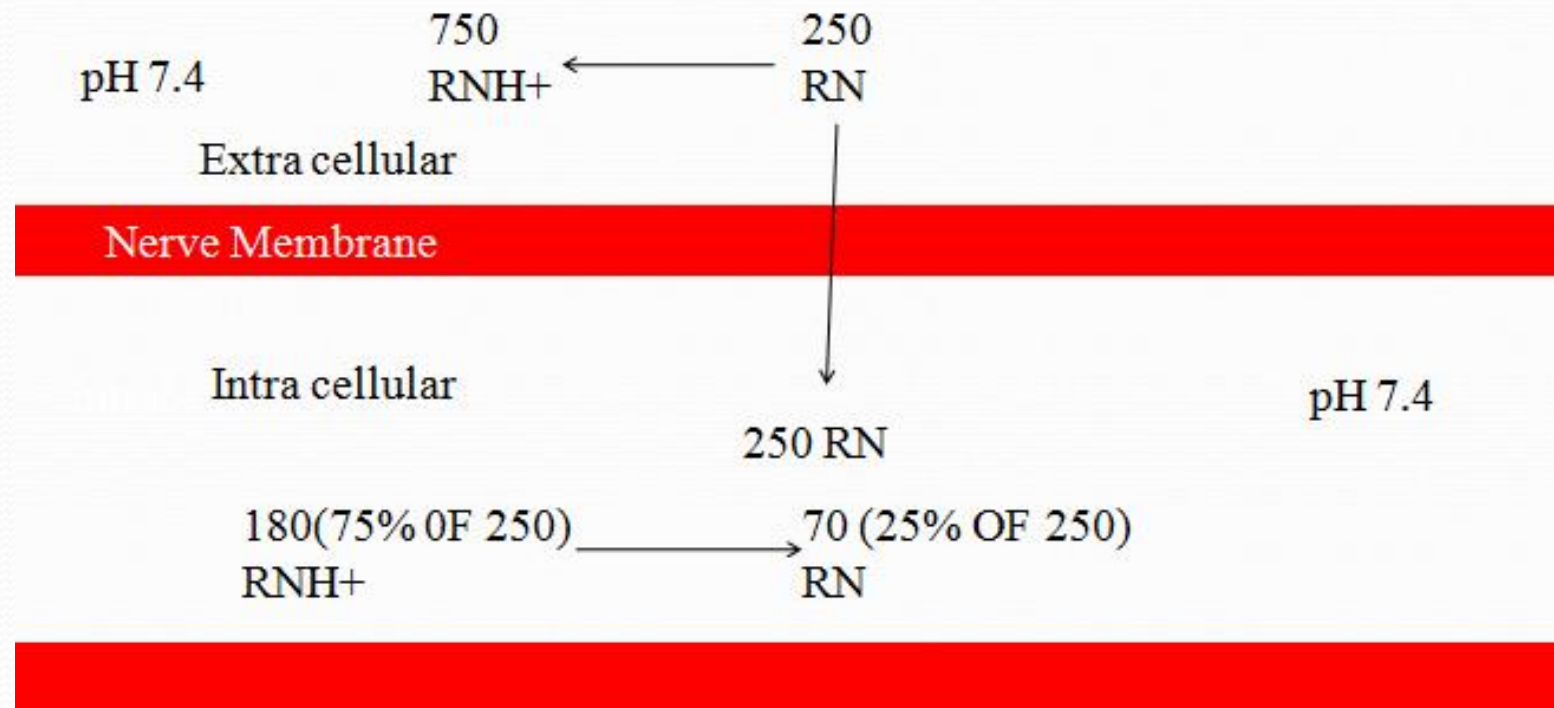
<u>Drug</u>	<u>pKa</u>	<u>RN Base Form</u>	<u>RNH⁺ Cation Form</u>
Mepivacaine-----	7.7-----	33%	67%
Lidocaine-----	7.7-----	33%	67%
Bupivacaine-----	8.1-----	17%	83%

TABLE 1-4**Dissociation Constants (pK_a) of Local Anesthetics**

Agent	pK_a	Percent Base (RN) at pH 7.4	Approximate Onset of Action (min)
Benzocaine	3.5	100	—
Mepivacaine	7.7	33	2 to 4
Lidocaine	7.7	29	2 to 4
Prilocaine	7.7	25	2 to 4
Articaine	7.8	29	2 to 4
Etidocaine	7.9	25	2 to 4
Ropivacaine	8.1	17	2 to 4
Bupivacaine	8.1	17	5 to 8
Tetracaine	8.6	7	10 to 15
Cocaine	8.6	7	—
Chloroprocaine	8.7	6	6 to 12
Propoxycaine	8.9	4	9 to 14
Procaine	9.1	2	14 to 18
Procainamide	9.3	1	—

Example for 1000 molecules of Etidocaine drug ($pK_a=7.9$)

With result of Henderson Hesselbach Equation Etidocaine has
25% in Base form RN & 75% in Cation form RNH^+ at 7.4 pH



Only 250 RN Base form molecules has the capacity to cross the nerve membrane (phospholipid layer) as they are lipophilic.

Once these 250 molecules gets entry into the nerve there are no RN Base form molecules left outside the nerve.

750 Molecules of RNH^+ which are remaining outside the nerve will get interconverted to RN as per Henderson-Hasselbalch Equation

25% of 750 = 180 Molecules in RN Base form

75% of 750 = 570 molecules in RNH^+ Cation form

Again 180 RN Base form will get entry in to the nerve.....These 180 RNH^+ Cation form molecules attach to the receptors in the Sodium Channels and block the entry of sodium ions.

- Local Anesthetics with higher pka value > 7.5 pKa

Have **Slow onset of Local Anesthetic action**. This is because of less number of RN Base form molecules at the body pH 7.4

- Local Anesthetics with lower pka value < 7.5 pka

Have **early onset of Local Anesthetic action**. This is because of more number of RN Base form molecules at the body pH 7.4

What happens when the infection is present in the tissues

- Infection in the tissues will lead to acidic pH in the tissues. This is because of **more hydrogen H^+ ions** present in the infected tissues.
- The normal pH of the human body is 7.4. But in the infection areas the pH will be more acidic 5 to 6 (More H^+ ions)
- So the pH will be less than 7.4 in these infected areas.

- When Local Anesthetic Solution is deposited in the infected tissues there will be more **RNH⁺ Cation molecules** and less RN Base molecules.
- Only these less RN Base form molecules cross the nerve membrane and get inside the nerve.
- So overall efficiency of the drug is reduced in infected tissues.

Same example of Etidocaine but during injection at acidic Ph of 6

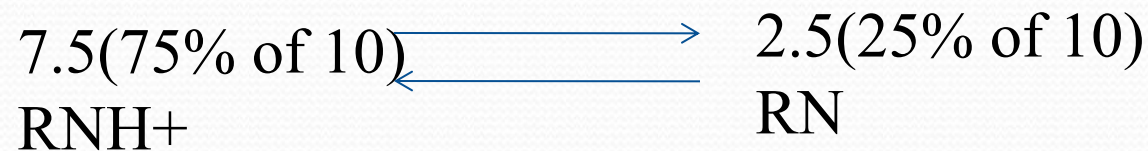
Extracellular Ph 6 (Acidic ph)



Nerve Membrane

Intracellular pH 7.4

10 RN



pH and LA

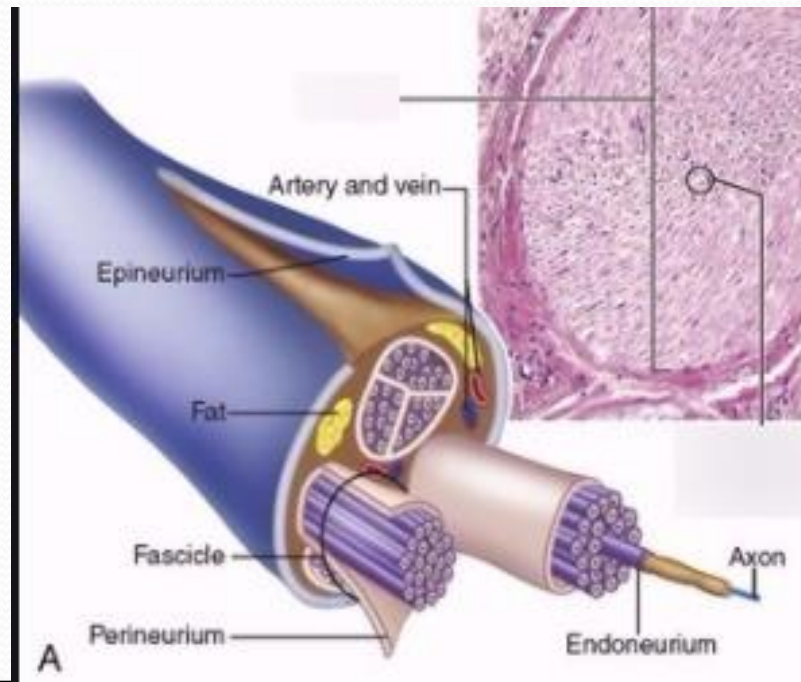
The pH of a LA solution and the pH of the tissue into which it is deposited greatly influences its nerve-blocking action.

- The pH of normal tissue is 7.4.
- **The pH of an inflamed area is 5-6 (acidic).**
- Acidification of tissue decreases LA effectiveness.
- Inadequate anesthesia results when LAs are injected into infected or inflamed area.
- The inflammatory process produces acidic products.
- Some LAs contain **epinephrine** , a vasoconstrictor which is highly unstable, Oxidize and turns brown.
- Sodium bisulfite, **an antioxidant** is added to inhibit the oxidation of Epinephrine. This makes the solution more acidic.

Kinetics of Local Anesthetic Agents

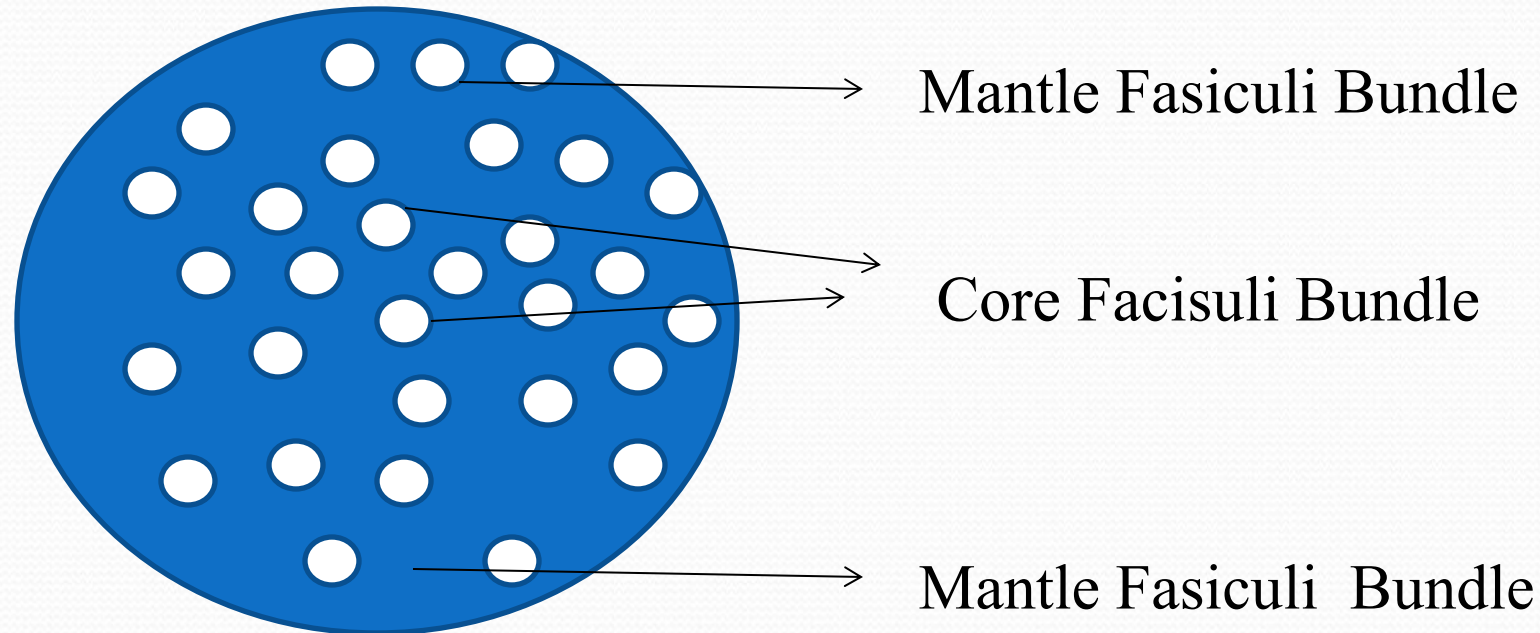
Barrier to diffusion of the solution

- Local Anesthetic solution after deposition in to the tissue tries to spread by diffusion.
- **Perineurium** is the greatest barrier for the diffusion of the local anesthetic molecules



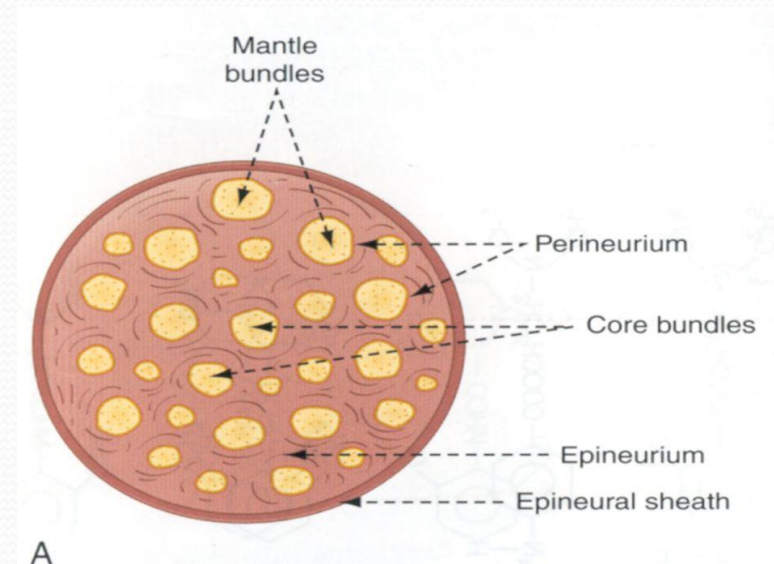
Mantle Bundles & Core Bundles

Cross section of Nerve



Mantle fibres tend to innervate more proximal areas, such as molars area with inferior alveolar nerve block, core fibers innervate incisors and canine.

- Local anesthetic solution reaches Mantle Fasculi bundle first and Core Fasculi bundles at the end.
- By the time the solution reaches the core bundle area the concentration of the drug will be reduced.
- So the effect of anesthesia in the core bundle is at the end, and this is the reason the effect in core bundle is not profound (not stronger).



Diffusion of the Local Anesthetic drug after deposition in tissue

- Solution diffuses in all direction in tissue.
- Some part of the drug will cross the nerve membrane
- Some drug is absorbed by non neuronal tissue like fat & muscle
- Some part of drug is diluted by body fluid
- Some part of drug is absorbed by Blood vessels & lymphatic drainage
- If ester type of local anesthetic drug - then it is hydrolysed immediately in the plasma.

Induction time:

It is defined as the time required from the deposition of local anesthetic solution till the conduction blockade.

Factors which influence the effect of local anesthetic Agents are

1. **Concentration of the drug –**

Greater the Concentration of the drug means more molecules are present hence more potent(stronger) the action

Example ; Lignocaine 2% means 20mg in 1 ml of solution

Lignocaine 3% means 30 mg in 1ml of solution

2. **Anatomic barrier**

Perineurium is the greatest barrier for the diffusion of the Local Anesthetic agents

3. **Dissociation Constant (pKa)**

Drug with Higher pKa Number ----- Slower the onset of action

Drug with Lower the pKa number ----- Faster the onset of action

4. **Lipid solubility**

- Greater the Lipid solubility of drug, the more potent the drug is.

- Lesser the Lipid solubility of the drug , less potent the drug is

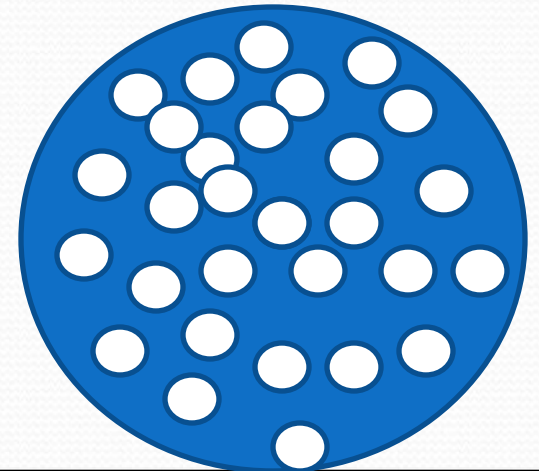
5. **Protein binding capacity**

- Greater the degree of protein binding , Longer the duration of action.

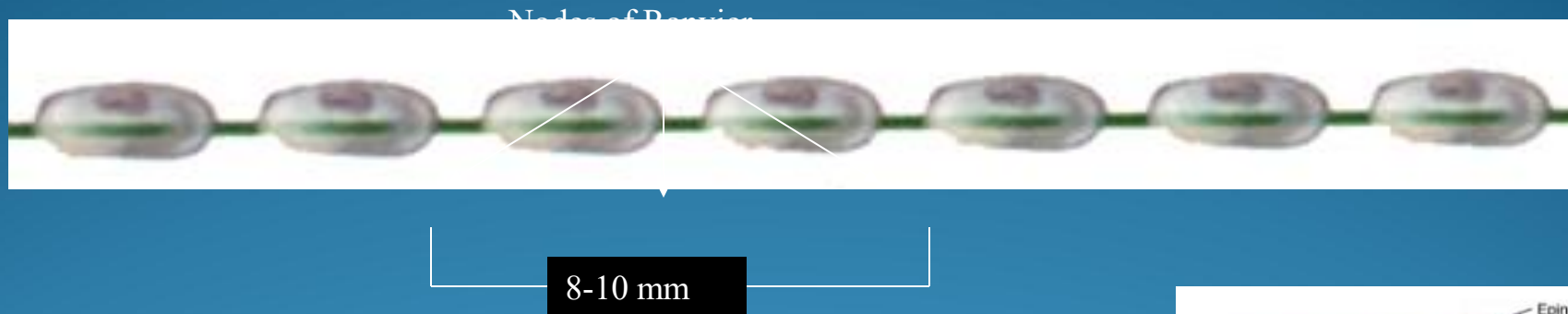
- Lesser the degree of protein binding , Shorter the duration of action

Recovery from the Local Anaesthetic Block

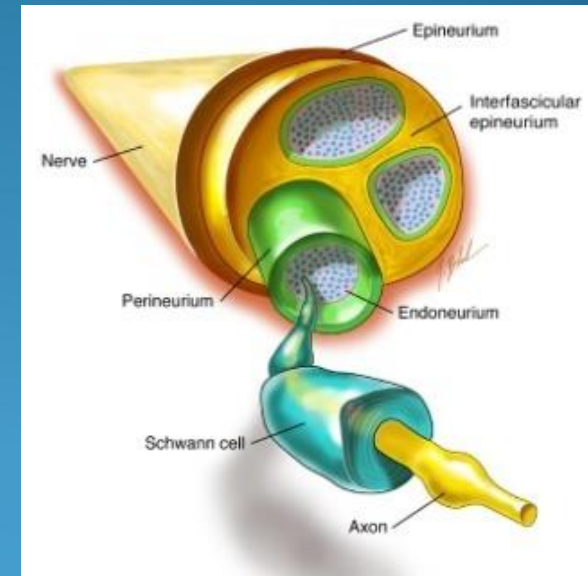
- Extra neural concentration of local anaesthetic agent is reduced by diffusion , dispersion, and uptake from the blood vessel.
- Concentration gradient is reversed . Fasculi in the mantle area lose the local anaesthetic drug first then Fasculi of the core region.
- Local anesthetic within core area diffuses to Mantle area .
- First area to lose anaesthetic effect is core area fibers.



Factors Governing Maximum Effect of LA



- Area of nerve in contact with the drug
8-10mm – 2-3 nodes of Ranvier should be blocked
- Size of the nerve
The larger the nerve the more difficult to blocked



Tachyphylaxis

- An acute, sudden decrease in response to a drug after its administration
- a rapid and short-term onset of drug tolerance can occur after repeated doses of reinjection.
- The duration, intensity of reinjection will be greatly reduced.
- It can be due to edema, decreased PH, localized hemorrhage and clot formation.

Thank You

References :

Page No 11-23, Chapter-1 Neurophysiology

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7th Edition

- "Nerve Impulse Molecular Mechanism [3D Animation]" on YouTube
<https://youtu.be/fHRC8SILcHo?si=cYlQyFg6MphXGqwO>